NumPy Array Iterating

Iterating Arrays

```
import numpy as np

arr = np.array([1, 2, 3])

for x in arr:
  print(x)
```

```
1
2
3
```

Iterating 2-D Arrays

```
import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]])

for x in arr:
  print(x)
```

```
[1 2 3]
[4 5 6]
```

Iterate on each scalar element of the 2-D array:

```
import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]])

for x in arr:
  for y in x:
    print(y)
```

```
1
2
3
4
5
6
```

Iterating 3-D Arrays

```
import numpy as np

arr = np.array([[[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]]])
```

```
for x in arr:
  print(x)
```

```
[[1 2 3]
 [4 5 6]]
[[ 7  8  9]
 [10 11 12]]
```

Iterate down to the scalars:

```
import numpy as np

arr = np.array([[[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]]])

for x in arr:
  for y in x:
    for z in y:
      print(z)
```

```
1
2
3
4
5
6
7
8
9
10
11
12
```

Iterating Arrays Using nditer()

```
import numpy as np

arr = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])

for x in np.nditer(arr):
  print(x)
```

```
1
2
3
4
5
6
7
8
```

Iterating Array With Different Data Types

```
import numpy as np

arr = np.array([1, 2, 3])

for x in np.nditer(arr, flags=['buffered'], op_dtypes=['S']):
  print(x)
```

```
np.bytes_(b'1')
np.bytes_(b'2')
np.bytes_(b'3')
```

Iterating With Different Step Size

```
import numpy as np

arr = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])

for x in np.nditer(arr[:, ::2]):
  print(x)
```

```
1
3
5
7
```

Enumerated Iteration Using ndenumerate()

```
import numpy as np

arr = np.array([1, 2, 3])

for idx, x in np.ndenumerate(arr):
  print(idx, x)
```

```
(0,) 1
(1,) 2
(2,) 3
```

Enumerate on following 2D array's elements:

```
import numpy as np

arr = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])

for idx, x in np.ndenumerate(arr):
  print(idx, x)
```

```
(0, 0) 1
(0, 1) 2
(0, 2) 3
(0, 3) 4
(1, 0) 5
(1, 1) 6
```

```
(1, 2) 7
(1, 3) 8
```